

AjJay Kannan

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Summary

Data scientist and Applied AI engineer who ships production AI across the full stack spanning classical ML, optimization, LLM systems, and agent platforms, for supply chain and enterprise operations. Over 3+ years at CHS: built a substitute-product recommender using LLM attribute extraction and embedding similarity that lifted in-full-on-time delivery by 16%, an LLM-agent code-review system that cut issue-detection time by 95%, and end-to-end data platform and workflow automation initiatives that eliminated 1,000+ manual hours/month. Outside CHS, built a substantial self-directed AI portfolio spanning agent platforms, on-device multimodal LLM apps, and a PyPI-published Microsoft MarkItDown plugin.

Skills

Methods RAG, Document Intelligence, Time-Series Forecasting, Optimization, Agent Harness, A/B Testing

LLM Agent SDK, LangGraph, MCP, Edge AI (LiteRT-LM, Whisper.cpp)

DevOps/IaC Terraform, GitHub Actions, Azure Pipelines, Docker

Languages Python, SQL, R, Bash

Platform Snowflake, dbt, Confluent, AWS, Azure

ML scikit-learn, XGBoost, LightGBM, Prophet

Databases Postgres, MongoDB, Qdrant

Deep Learning PyTorch, TensorFlow, LSTM

BI Power BI, Tableau

Projects

Supply-Chain Planning Agent Platform

- Building a **kernel + sub-agents** platform that scaffolds the judgment around demand planning: a Decision Memory Agent captures planner-override rationale at commit time, and a Grey Swan Sentinel tracks political, economic, and environmental signals hourly to push override recommendations back into the workflow
- Stack: **Claude Agent SDK**, **MCP**, **Next.js**, **Supabase Postgres + pgvector**, Drizzle ORM with RLS-aware migrations

Financial Filings Research Agent [🔗](#)

- Multi-agent research workspace for corporate filings — pulls BSE India announcements, runs structured extraction and hybrid retrieval over PDFs, and exposes every conversation thread as an isolated workspace with its own memory, notebook, and staged documents
- Built on the **Claude Agent SDK** with a dedicated **agent harness** (MCP tools, skills, hooks); stack includes Next.js 16, TypeScript, OpenAI, Cosmos DB

Inside Dreamer — Beta-Tester Architecture Writeup [🔗](#)

- Long-form writeup of agent-platform architecture from beta-testing Dreamer (now part of **Meta Superintelligence Labs**) and shipping multiple agents end-to-end on the platform during its beta program
- Covers **kernel + sub-agent VM** composition, typed function exports stitched at runtime, and persistent memory as a platform primitive, patterns now shaping the agent platforms I build

WhereIKept — On-Device Multimodal Android Assistant [🔗](#)

- Android app that runs lightweight multimodal LLMs **entirely on-device** — capture a voice note and photo while putting something away, later search in natural language for where you stored it
- **Whisper.cpp** for speech-to-text and **Gemma 3N E2B** multimodal LLM on Google's **LiteRT-LM** runtime, all local; **Room FTS4** powers search, HuggingFace OAuth + WorkManager handles gated-model background downloads

Microsoft/MarkItDown PDF Plugin [🔗](#)

- Open-source plugin published to **PyPI** for Microsoft's MarkItDown library; converts complex multi-modal PDFs (tables, figures, layout) into LLM-ready Markdown for stronger RAG ingestion, processing pages concurrently

Kronia + EPA Ingestion Pipeline [🔗](#)

- Six-stage production-style ingestion pipeline for RAG (API ingest → Snowflake landing → **CORTEX.PARSE.DOCUMENT** → chunking UDF → **Cortex Search Service**) feeding a multimodal chat over EPA pesticide labels with weather context
- Containerized on **AWS Fargate** via **Terraform** with daily **EventBridge** scheduling and CI/CD via **GitHub Actions**; chat shipped on Streamlit for the Snowflake × Mistral Hackathon

Experience

CHS Inc

Data Scientist — Supply Chain Planning and Execution

Previously: Data Analyst Intern

St. Paul, MN

Jun 2023 – Present

Jan 2023 – Apr 2023

- Led **end-to-end design, architecture, and deployment of scalable data and AI products** for supply chain and operations, partnering with cross-functional teams to drive **production adoption and change management**
- Engineered an **LLM-driven documentation workflow** for data products, cutting manual effort by **~40 hours per week**
- Developed an **automated code-review system** using **LLM agents** that reduced issue detection time by **95%**
- Designed and deployed a **substitute-product recommendation system** through extracting key attributes from large **product-label PDFs** using LLM and using **embedding similarity**; improved search efficiency by **85%** and increased **SIFOT** by **16%**
- Built and supported **demand forecasting and planning application** using statistical and ML models (**Prophet, LSTM**), reducing planning cycle time by **20%** and improving forecast accuracy by **9%**
- Built a **mixed-integer optimization system** for inventory assignment and blending under **quality and quantity constraints**. Enabled **order-feasibility decisions** and eliminated manual blending workflows, saving **30 hours/week**
- Built scalable **Snowflake/dbt data products** for inventory and terminal operations, enabling **60-day stockout visibility** and **capacity-utilization KPIs** across **50+ terminals**. Eliminated **34 documents** and saved **~750 hours/month**

Texas A&M University

Data Analyst — Student Technician (Part Time)

College Station, TX

Sep 2021 – Jan 2023

- Reduced churn from **35%** to **20%** in the Engineering Honors Program by conducting **Cohort Analysis** in Tableau, applying **K-Means Clustering** for Segmentation, and performing **Hypothesis Testing**

o9 Solutions Inc

Business Operations Specialist Intern

Dallas, TX

Jun 2022 – Aug 2022

- Developed **Python Plugins** for data integration using **ETL** processes in **Apache Spark** and optimized SQL queries to enhance models for Supply Chain Planning and Control Tower Functionalities

Tata Motors Limited

Operations Manager

Previously: Graduate Engineer Trainee

Pune, India

Aug 2019 – Jun 2021

Aug 2018 – Jul 2019

- Led a 50+ member cross-functional team to deliver 13 CKD vehicle projects, **implementing lean and ML-driven tools** while achieving 100% fulfillment, 23% lower stock-out costs, 4% station downtime, and 30% faster claims closure

Education

Texas A&M University

Master of Science in Industrial Engineering (GPA: 3.89/4.00)

Aug 2021 – May 2023

PSG College of Technology

Bachelor of Engineering in Production Engineering (GPA: 8.61/10.00)

Jun 2014 – June 2018